Branch Specialization Checkpoint

Explicit Branch Isolation

Branch Specialization Project

2026-05-22

Checkpoint reason: explicit routing separates specialization from modularity.

Question

Previous toy results showed:

- head-dimension heterogeneity can stabilize functional specialization;
- adding a second high-capacity head does not force modular role separation.

This checkpoint asks:

Is explicit branch routing needed to turn specialization into functional modularity?

Setup

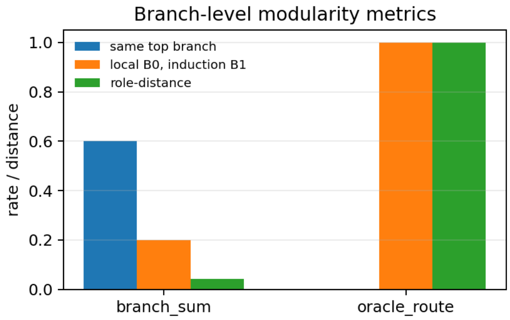
Same local-vs-induction competition task:

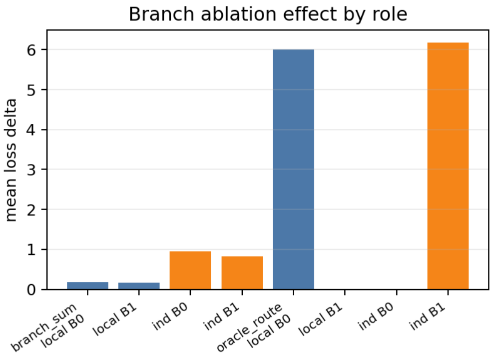
- local copy: `[x, SEP, x]`;
- induction continuation: `[y1,...,y16,y1,...,y16]`;
- local weight 0.25, induction weight 1.0;
- five seeds per config.

Branch variants:

- `branch_sum`: two separate branch towers, both active everywhere.
- `oracle_route`: local positions use branch 0; induction positions use branch 1.

Main Evidence





Ungated branches stay entangled. Oracle routing produces role-specific branch ablations.

Summary Table

Config	Local acc.	Ind. acc.	Same top branch	Routed match	Branch distance
`branch_sum`	1.0000	0.9998	0.60	0.20	0.0411
`oracle_route`	0.9999	0.9987	0.00	1.00	1.0000

Branch distance is the difference between local and induction branch-specialization distributions. Higher means stronger modular separation.

Branch Ablation

Config	Local B0	Local B1	Induction B0	Induction B1
`branch_sum`	0.1860	0.1672	0.9538	0.8308
`oracle_route`	6.0104	0.0000	0.0000	6.1801

In `branch_sum`, both branches support both roles. In `oracle_route`, branch 0 is local-specific and branch 1 is induction-specific.

Interpretation

Supported:

- Explicit routing can produce functional modularity in this toy setting.
- Separate branch towers alone do not discover modular roles.
- Specialization and modularity are experimentally separable.

Current project claim:

> *Capacity heterogeneity stabilizes specialization; routing or stronger structural separation is needed for robust modularity.*

Decision

Continue Phase 3 with two distinct claims:

1. structural heterogeneity can stabilize specialization;
2. explicit routing/separation can produce modularity.

Next decisive test:

- replace oracle routing with learned or weakly supervised routing;
- test whether modularity emerges without hard-coded task-region assignment.

Sources and Provenance

- Report: `doc/phase3_toy_branch_isolation.md`
- Model script: `scripts/toy_branch_isolation_intervention.py`
- Analysis script: `scripts/analyze_branch_isolation.py`
- Run output: `results/phase3_toy_branch_isolation_lw025/`
- Analysis output: `results/phase3_toy_branch_isolation_analysis/`
- Deck data snapshot: `presentations/2026-05-22-1230-branch-isolation/data/`